1.Creation of Amazon EC2 Instances (Linux and Windows) and Establishing Secure Connections Using SSH from Local Machine, PuTTY (.pem & .ppk), AWS CloudShell, and Remote Desktop Protocol (RDP)

🗒️ **PREREQUISITES**

Make sure you have:

- ✅ EC2 **Ubuntu** instance running
- ✅ **.ppk private key file**
- ✅ **PuTTY installed (Windows)-- https://putty.org/index.html (INSTALL FIRST ONE)**

🔹 **STEP 1: Check Security Group**

Go to **EC2 → Security Groups → Inbound rules**

Ensure this rule exists:

Type: SSH

Port: 22

Source: My IP

❌ If SSH is blocked → connection will fail

🔹 **STEP 2: Open PuTTY**

1. Launch **PuTTY**
2. In **Session** screen:

Host Name (or IP address): EC2—CONNECT—COPY (DNS NAME)

ubuntu@<PUBLIC-IP>

Example:

ubuntu@3.110.xxx.xxx

🔹 STEP 3: Attach the .ppk Key

In PuTTY left menu, go to:

Connection → SSH → Auth → Credentials

1. Click **Browse**
2. Select your **.ppk file**
3. Click **Open**

🔹 STEP 4: Connect to EC2

Click **Open**

- First time → click **Yes** (security alert)
- PuTTY will log in automatically as:

ubuntu

🎉 **You are now connected to your Ubuntu EC2 instance**

🧪 VERIFY CONNECTION

Run:

whoami

Output should be:

Ubuntu

Connect EC2 Ubuntu Instance Using PuTTY (Windows) ---USING PEM KEYPAIR

PREREQUISITES

You must have:

- ✅ **EC2 Ubuntu instance running**
- ✅ **Public IPv4 address** (or Elastic IP)
- ✅ **Key pair (.pem file)** used while launching EC2
- ✅ **PuTTY & PuTTYgen installed**

👉 Download PuTTY:
https://www.putty.org

🔹 **STEP 1: Convert .pem to .ppk using PuTTYgen**

PuTTY **cannot use .pem directly**.

Steps:

1. Open **PuTTYgen**
2. Click **Load**
3. Select your .pem file
   (change file type to *All Files*)
4. Click **Open**
5. Click **Save private key**
6. Save as keyname.ppk

✓ Conversion done

🔹 **STEP 2: Get EC2 Public IP**

Go to:
AWS Console → EC2 → Instances---CONNECT –COPY DNS NAME

🔹 **STEP 3: Configure PuTTY**

1️⃣ **Open PuTTY**

Session:

- **Host Name (or IP address): dns name**

2️⃣ **Load Private Key**

Go to:

Connection → SSH → Auth → Credentials

- Click **Browse**
- Select your .ppk file

🔹 **STEP 4: Connect**

Click **Open**

- First time → click **Yes** (security alert)
- Login happens automatically as:

ubuntu

🎉 **You are now connected to Ubuntu EC2**

🧪 **VERIFY CONNECTION**

Run:

whoami

Output:

ubuntu

):

Connecting to EC2 Using AWS CloudShell

1. **Create the EC2 instance** using a key pair (.pem).
2. **Log in to AWS Console** and click on the **CloudShell** icon (terminal icon on the top-right).

3. Once CloudShell opens, go to **AWS Services → EC2**.
4. Select your **EC2 instance** and click **Connect**.
5. Choose the **SSH client** tab.
6. **Copy the ssh -i command** shown there.
7. **Paste the command in CloudShell** and try to run it.
   ❌ It will fail with an error like:
   "key pair is not available / no such file or directory"
   (because the key file is not yet in CloudShell).
8. Now, in **CloudShell**, click on the **"+" (Actions)** icon on the right side.
9. Select **Upload file** and upload the **key pair (.pem)** from your local machine.
10. After upload, **copy the ssh -i command again** from **EC2 → Connect → SSH client**.
11. **Paste and execute the command** in CloudShell.
12. ✅ You are now **successfully connected to the EC2 instance**.

Step-by-Step: Create Windows EC2 Instance and Connect via RDP

Step 1: Launch a Windows EC2 Instance

1. Log in to your **AWS Management Console**.
2. Navigate to **EC2 → Instances → Launch Instances**.
3. Select an **Amazon Machine Image (AMI)**:
   - Choose a **Windows Server** AMI (e.g., Windows Server 2019 or 2022).
4. Choose an **Instance Type** (e.g., t2.micro for free tier or as per your requirement).
5. Configure **Instance Details** as needed.
6. Add **Storage** if required (default is usually sufficient).
7. Configure **Security Group**:
   - Ensure **RDP port 3389** is allowed.
   - Source can be **My IP** (for security) or a wider range if needed.

8. Select or **create a new Key Pair**:
    - Download the .pem file (keep it safe; it's needed to decrypt the password).
9. Click **Launch Instance**.

Step 2: Get the Public DNS / IP

1. After the instance is running, go to **EC2 → Instances**.
2. Select your instance and note the **Public DNS (IPv4)** or **Public IP**.

Step 3: Retrieve the Windows Administrator Password

1. Select your instance → Click **Connect → RDP Client**.
2. Click **Get Password**.
3. Upload your **Key Pair (.pem) file**.
4. Click **Decrypt Password**.
5. Copy the **Administrator password** shown.

Step 4: Connect via RDP

1. On your **local Windows machine**, open **Remote Desktop Connection** (mstsc).
2. In the **Computer** field, enter the **Public DNS** or **IP address** of the EC2 instance.
3. Click **Connect**.
4. In the login prompt:
    - **Username**: Administrator
    - **Password**: Paste the decrypted password from Step 3.
5. Click **OK**.

Step 5: You're connected

- Your Windows EC2 instance desktop should appear.
- You can now use it as a normal Windows machine.

Scenario-Based Questions on EC2 Creation & Secure Access

1. You are launching a **Linux EC2 instance** for the first time. AWS asks you to select a **key pair**.
   Why is a key pair mandatory, and what will happen if you skip it?
2. You created an EC2 instance, but **SSH connection times out** from your local machine.
   What security group–related issues would you check first?
3. You want to allow **only your office IP** to connect to EC2 via SSH.
   How would you configure the security group, and why is this important?
4. You launched a **Windows EC2 instance** and want to connect using **RDP**.
   Which port must be opened, and why should it not be open to 0.0.0.0/0?
5. You are working on a **Windows laptop** and want to connect to a Linux EC2 instance using **PuTTY**.
   Why do you need to convert a .pem file to .ppk?
6. A Linux EC2 instance was launched using an **Amazon Linux AMI**.
   What does an AMI contain, and how does it affect instance behavior?
7. You need to create **multiple identical EC2 instances** with the same software installed.
8. You want to connect to EC2 but **cannot use your local machine** due to corporate restrictions.
9. You accidentally deleted the **inbound SSH rule** from a security group. **What immediate impact will this have on existing EC2 connections?**
10. You are creating a **production EC2 instance**. **Why should you avoid using the default security group?**
11. A Windows EC2 instance is running, but **RDP login fails** even though the instance is healthy.
    What key pair–related or configuration issues could cause this?
12. You want to automate EC2 creation using the **same OS, disk size, and software stack**.
    Which EC2 component helps you standardize this setup?
13. You launched an EC2 instance using the wrong **AMI (Windows instead of Linux)**.
    Can you change the AMI after launch? What would you do instead?
14. You created a security group allowing SSH from anywhere for testing.
    Why is this dangerous in production, and how would you fix it?

15. How do you to migrate an EC2 setup from **one region to another**.